STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE # 23

STATISTICAL INFERENCE

INFERENCE ABOUT MEAN

PREPARED BY
HAZBER SAMSON
FAST-NUCES ISLAMABAD

INFERENCES ON A SINGLE POPULATION

INFERENCE ABOUT THE POPULATION MEAN WHEN SIGMA IS KNOWN

CASE-1 $If \;\; H_0 : \mu = \mu_\circ \;\; vs \;\; H_1 : \mu \neq \mu_\circ$

- When σ is known then we use **test statistic**

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- When σ is known then we use **P-Value**

$$P = 2[1 - \Phi(|z_0|)]$$

- The $(1-\alpha)\ 100\%$ **confidence interval** for μ is

$$\bar{x} - Z_{\frac{\alpha}{2}} . \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\frac{\alpha}{2}} . \frac{\sigma}{\sqrt{n}}$$

CASE-2 $If \;\; H_0 : \mu \geq \mu_\circ \;\; vs \;\; H_1 : \mu < \mu_\circ$

- When σ is known then we use **test statistic**

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- When σ is known then we use **P-Value**

$$P = \Phi(z_0)$$

- The $(1-\alpha)\ 100\%$ **confidence interval** for μ is

$$\mu \leq \bar{x} + Z_\alpha . \frac{\sigma}{\sqrt{n}}$$

CASE-3 $If \;\; H_0 : \mu \leq \mu_0 \;\; vs \;\; H_1 : \mu > \mu_\circ$

- When σ is known then we use **test statistic**

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- When σ is known then we use **P-Value**

$$P = 1 - \Phi(z_0)$$

- The $(1-\alpha)\ 100\%$ **confidence interval** for μ is

$$\mu \geq \bar{x} - Z_\alpha . \frac{\sigma}{\sqrt{n}}$$

HAZBER SAMSON

Z-TEST

If $H_\circ : (=, \leq, \geq)$ then $H_1 : (\neq, >, <)$

CRITICAL REGIONS IN Z-TEST

- If H_1 *contains* $\neq$ then C.R is $|z| \geq Z_{\frac{\alpha}{2}}$ (Two Sided)
- If H_1 *contains* $>$ then C.R is $Z > Z_\alpha$ (One Sided)
- If H_1 *contains* $<$ then C.R is $Z < -Z_\alpha$ (One Sided)

FREQUENTLY USED CRITICAL VALUES OF Z

Significance Level (α)	Confidence Level $(1-\alpha)$	One-Tailed Test	Two-Tailed Test
$\alpha = 0.01$	99%	$Z_\alpha = 2.33$	$Z_{\alpha/2} = 2.58$
$\alpha = 0.02$	98%	$Z_\alpha = 2.05$	$Z_{\alpha/2} = 2.33$
$\alpha = 0.03$	97%	$Z_\alpha = 1.88$	$Z_{\alpha/2} = 2.17$
$\alpha = 0.04$	96%	$Z_\alpha = 1.75$	$Z_{\alpha/2} = 2.05$
$\alpha = 0.05$	95%	$Z_\alpha = 1.65$	$Z_{\alpha/2} = 1.96$
$\alpha = 0.06$	94%	$Z_\alpha = 1.55$	$Z_{\alpha/2} = 1.88$
$\alpha = 0.07$	93%	$Z_\alpha = 1.48$	$Z_{\alpha/2} = 1.81$
$\alpha = 0.08$	92%	$Z_\alpha = 1.41$	$Z_{\alpha/2} = 1.75$
$\alpha = 0.09$	91%	$Z_\alpha = 1.34$	$Z_{\alpha/2} = 1.70$
$\alpha = 0.10$	90%	$Z_\alpha = 1.28$	$Z_{\alpha/2} = 1.65$

P-VALUE FORMULAS IN Z-TEST

$$P = \begin{cases} 2[1-\Phi(|z_0|)] & \text{if} \quad H_1 : \mu \neq \mu_0 \\ 1-\Phi(z_0) & \text{if} \quad H_1 : \mu > \mu_0 \\ \Phi(z_0) & \text{if} \quad H_1 : \mu < \mu_0 \end{cases}$$

HAZBER SAMSON

EXAMPLES OF INFERENCE ABOUT MEAN WHEN SIGMA IS KNOWN

EXAMPLE-1 A researcher claims that the average wind speed in a certain city is 8 miles per hour. A sample of 32 days has an average wind speed of 8.2 miles per hour. The standard deviation of the population is 0.6 mile per hour. At $\alpha = 0.05$, is there enough evidence to reject the claim? Use all the three approaches to verify the claim.

SOLUTION *Here* $n = 32,\ \bar{x} = 8.2,\ \sigma = 0.6,\ \alpha = 0.05$

CRITICAL VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$$H_0 : \mu = 8$$

$$H_1 : \mu \neq 8$$

Step -2 : *Level of Significance*

$$\alpha = 0.05$$

Step -3 : *Test Statistic*

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{8.2 - 8}{0.6/\sqrt{32}} = 1.89$$

Step -4 : *Critical* Re *gion*

$$|z| \geq z_{\frac{\alpha}{2}} \quad \Rightarrow |z| \geq 1.96$$

Step -5 : *Conclusion*

Since calculated value of z does not lies in CR so do not reject H_0.

P-VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$$H_0 : \mu = 8$$

$$H_1 : \mu \neq 8$$

Step -2 : *Level of Significance*

$$\alpha = 0.05$$

Step -3 : *Test Statistic*

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{8.2 - 8}{0.6/\sqrt{32}} = 1.89$$

Step -4 : $P - value$

$$p = 2[1 - \Phi(|z_0|)] = 2[1 - \Phi(|1.89|)] = 0.0588$$

Step -5 : *Conclusion*

Since $p - value > \alpha$ *so do not reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1 : *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\bar{x} - Z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

$$8.2 - (1.96)(0.6)/\sqrt{32} \leq \mu \leq 8.2 - (1.96)(0.6)/\sqrt{32}$$

$$7.99 \leq \mu \leq 8.41$$

Step -2 : *Conclusion*

Since $\mu = 8$ *lies in CI so do not reject* H_0.

HAZBER SAMSON

EXAMPLE-2 A researcher wishes to test the claim that the average cost of tuition and fees at a four year public college is greater than \$5700. She selects a random sample of 36 four-year public colleges and finds the mean to be \$5950. The population standard deviation is \$659. Is there evidence to support the claim at $\alpha = 0.05$?

SOLUTION *Here* $n = 36,\ \bar{x} = 5950,\ \sigma = 659,\ \alpha = 0.05$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu \le \$5700$

$H_1 : \mu > \$5700$

Step -2: *Level of Significance*

$\alpha = 0.05$

Step -3: *Test Statistic*

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{5950 - 5700}{659/\sqrt{36}} = 2.28$$

Step -4: *Critical* Re *gion*

$z > z_\alpha \quad \Rightarrow z > 1.65$

Step -5: *Conclusion*

Since calculated value of z *lies in CR so reject* H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu \le \$5700$

$H_1 : \mu > \$5700$

Step -2: *Level of Significance*

$\alpha = 0.05$

Step -3: *Test Statistic*

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{5950 - 5700}{659/\sqrt{36}} = 2.28$$

Step -4: $P - value$

$p = 1 - \Phi(z_0) = 1 - \Phi(2.28) = 0.0113$

Step -5: *Conclusion*

Since $p - value < \alpha$ *so reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\mu \ge \bar{x} - Z_\alpha . \frac{\sigma}{\sqrt{n}}$$

$\mu \ge 5950 - (1.65)(659)/\sqrt{36}$

$\mu \ge 5768.78$

Step -2: *Conclusion*

Since $\mu = 5700$ *does not lie in CI so reject* H_0.

Summarize the results. There is enough evidence to support the claim that the tuition and fees at four-year public colleges are greater than \$5700.

HAZBER SAMSON

EXAMPLE-3 Rosie is an aging sheep dog in Montana who gets regular checkups from her owner, the local veterinarian. From past experience, the vet knows that Rosie's heart rate (beats per minute) has a normal distribution. The vet checked the Merck Veterinary Manual and found that for dogs of this breed $\mu = 115$ *and* $\sigma = 12$ beats per minute.

Over the past 6 weeks, Rosie's heart rate (beats/min) measured

93 109 110 89 112 117

The vet is concerned that Rosie's heaet rate may be slowing. Do the data indicate this is the case? Investigate using $\alpha = 0.01$.

SOLUTION *Here* $n = 6, \bar{x} = \sum x/n = 630/6 = 105,\ \sigma = 12, \alpha = 0.01$

CRITICAL VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \mu \geq 115$

$H_1 : \mu < 115$

Step -2 : *Level of Significance*

$\alpha = 0.01$

Step -3 : *Test Statistic*

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{105 - 115}{12/\sqrt{6}} = -2.04$$

Step -4 : *Critical* Re *gion*

$z < -z_\alpha \quad \Rightarrow z < -2.33$

Step -5 : *Conclusion*

Since calculated value of z does not lie in CR so do not reject H_0.

P-VALUE APPROACH

Step -1 : *Formulation of Hypotheses*

$H_0 : \mu \geq 115$

$H_1 : \mu < 115$

Step -2 : *Level of Significance*

$\alpha = 0.01$

Step -3 : *Test Statistic*

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{105 - 115}{12/\sqrt{6}} = -2.04$$

Step -4 : $P - value$

$p = \Phi(z_0) = \Phi(-2.04) = 0.0207$

Step -5 : *Conclusion*

Since $p - value > \alpha$ *so do not reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1 : *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\mu \leq \bar{x} + Z_\alpha . \frac{\sigma}{\sqrt{n}}$$

$\mu \leq 105 + (2.33)(12)/\sqrt{6}$

$\mu \leq 116.41$

Step -2 : *Conclusion*

Since $\mu = 115$ *lies in CI so do not reject* H_0.

HAZBER SAMSON

INFERENCES ON A SINGLE POPULATION

INFERENCE ABOUT THE POPULATION MEAN WHEN SIGMA IS UNKNOWN

CASE-1 $If \;\; H_0 : \mu = \mu_\circ \;\; vs \;\; H_1 : \mu \neq \mu_\circ$

- When σ is unknown then we use **test statistic**

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad with \quad d.f = n-1$$

- When σ is unknown then we use **P-Value**

$$P = 2 \times P(X \geq |t_0|)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\bar{x} - t_{\frac{\alpha}{2}(n-1)} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(n-1)} \cdot \frac{S}{\sqrt{n}}$$

CASE-2 $If \;\; H_0 : \mu \geq \mu_\circ \;\; vs \;\; H_1 : \mu < \mu_\circ$

- When σ is unknown then we use **test statistic**

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad with \; d.f = n-1$$

- When σ is unknown then we use **P-Value**

$$P = P(X \leq t_0)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\mu \leq \bar{x} + t_{\alpha(n-1)} \cdot \frac{S}{\sqrt{n}}$$

CASE-3 $If \;\; H_0 : \mu \leq \mu_0 \;\; vs \;\; H_1 : \mu > \mu_\circ$

- When σ is unknown then we use **test statistic**

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad with \quad d.f = n-1$$

- When σ is unknown then we use **P-Value**

$$P = P(X \geq t_0)$$

- The $(1-\alpha)$ 100% **confidence interval** for μ is

$$\mu \geq \bar{x} - t_{\alpha(n-1)} \cdot \frac{S}{\sqrt{n}}$$

HAZBER SAMSON

T-TEST

If $H_\circ : (=, \leq, \geq)$ then $H_1 : (\neq, >, <)$

Now for Critical Region

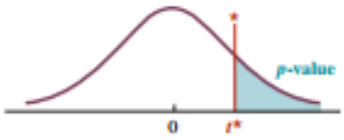
- If H_1 *contains* $\neq$ then C.R is $|t| \geq t_{\frac{\alpha}{2}(n-1)}$ (Two Sided)
- If H_1 *contains* $>$ then C.R is $t > t_{\alpha(n-1)}$ (One Sided)
- If H_1 *contains* $<$ then C.R is $t < -t_{\alpha(n-1)}$ (One Sided)

STUDENTS T-DISTRIBUTION TABLE

df	α = 0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.92	4.303	6.965	9.925	22.328	31.6
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.61
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.44	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.86	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.25	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.93	4.318
13	1.35	1.771	2.16	2.65	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.14
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.12	2.583	2.921	3.686	4.015
17	1.333	1.74	2.11	2.567	2.898	3.646	3.965
18	1.33	1.734	2.101	2.552	2.878	3.61	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.85
21	1.323	1.721	2.08	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.5	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.06	2.485	2.787	3.45	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.66
30	1.31	1.697	2.042	2.457	2.75	3.385	3.646
60	1.296	1.671	2	2.39	2.66	3.232	3.46
120	1.289	1.658	1.98	2.358	2.617	3.16	3.373
∞	1.282	1.645	1.96	2.326	2.576	3.091	3.291

Probability-Values for Student's *t*-distribution

The entries in this table are the *p*-values related to the right-hand tail for the calculated $t\star$ value for the *t*-distribution of df degrees of freedom.



	Degrees of Freedom														
$t\star$	3	4	5	6	7	8	10	12	15	18	21	25	29	35	df ≥ 45
0.0	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
0.1	0.463	0.463	0.462	0.462	0.462	0.461	0.461	0.461	0.461	0.461	0.461	0.461	0.461	0.460	0.460
0.2	0.427	0.426	0.425	0.424	0.424	0.423	0.423	0.422	0.422	0.422	0.422	0.422	0.421	0.421	0.421
0.3	0.392	0.390	0.388	0.387	0.386	0.386	0.385	0.385	0.384	0.384	0.384	0.383	0.383	0.383	0.383
0.4	0.358	0.355	0.353	0.352	0.351	0.350	0.349	0.348	0.347	0.347	0.347	0.346	0.346	0.346	0.346
0.5	0.326	0.322	0.319	0.317	0.316	0.315	0.314	0.313	0.312	0.312	0.311	0.311	0.310	0.310	0.310
0.6	0.295	0.290	0.287	0.285	0.284	0.283	0.281	0.280	0.279	0.278	0.277	0.277	0.277	0.276	0.276
0.7	0.267	0.261	0.258	0.255	0.253	0.252	0.250	0.249	0.247	0.246	0.246	0.245	0.245	0.244	0.244
0.8	0.241	0.234	0.230	0.227	0.225	0.223	0.221	0.220	0.218	0.217	0.216	0.216	0.215	0.215	0.214
0.9	0.217	0.210	0.205	0.201	0.199	0.197	0.195	0.193	0.191	0.190	0.189	0.188	0.188	0.187	0.186
1.0	0.196	0.187	0.182	0.178	0.175	0.173	0.170	0.169	0.167	0.165	0.164	0.163	0.163	0.162	0.161
1.1	0.176	0.167	0.161	0.157	0.154	0.152	0.149	0.146	0.144	0.143	0.142	0.141	0.140	0.139	0.139
1.2	0.158	0.148	0.142	0.138	0.135	0.132	0.129	0.127	0.124	0.123	0.122	0.121	0.120	0.119	0.118
1.3	0.142	0.132	0.125	0.121	0.117	0.115	0.111	0.109	0.107	0.105	0.104	0.103	0.102	0.101	0.100
1.4	0.128	0.117	0.110	0.106	0.102	0.100	0.096	0.093	0.091	0.089	0.088	0.087	0.086	0.085	0.084
1.5	0.115	0.104	0.097	0.092	0.089	0.086	0.082	0.080	0.077	0.075	0.074	0.073	0.072	0.071	0.070
1.6	0.104	0.092	0.085	0.080	0.077	0.074	0.070	0.068	0.065	0.064	0.062	0.061	0.060	0.059	0.058
1.7	0.094	0.082	0.075	0.070	0.066	0.064	0.060	0.057	0.055	0.053	0.052	0.051	0.050	0.049	0.048
1.8	0.085	0.073	0.066	0.061	0.057	0.055	0.051	0.049	0.046	0.044	0.043	0.042	0.041	0.040	0.039
1.9	0.077	0.065	0.058	0.053	0.050	0.047	0.043	0.041	0.038	0.037	0.036	0.035	0.034	0.033	0.032
2.0	0.070	0.058	0.051	0.046	0.043	0.040	0.037	0.034	0.032	0.030	0.029	0.028	0.027	0.027	0.026
2.1	0.063	0.052	0.045	0.040	0.037	0.034	0.031	0.029	0.027	0.025	0.024	0.023	0.022	0.022	0.021
2.2	0.058	0.046	0.040	0.035	0.032	0.029	0.026	0.024	0.022	0.021	0.020	0.019	0.018	0.017	0.016
2.3	0.052	0.041	0.035	0.031	0.027	0.025	0.022	0.020	0.018	0.017	0.016	0.015	0.014	0.014	0.013
2.4	0.048	0.037	0.031	0.027	0.024	0.022	0.019	0.017	0.015	0.014	0.013	0.012	0.012	0.011	0.010
2.5	0.044	0.033	0.027	0.023	0.020	0.018	0.016	0.014	0.012	0.011	0.010	0.010	0.009	0.009	0.008
2.6	0.040	0.030	0.024	0.020	0.018	0.016	0.013	0.012	0.010	0.009	0.008	0.008	0.007	0.007	0.006
2.7	0.037	0.027	0.021	0.018	0.015	0.014	0.011	0.010	0.008	0.007	0.007	0.006	0.006	0.005	0.005
2.8	0.034	0.024	0.019	0.016	0.013	0.012	0.009	0.008	0.007	0.006	0.005	0.005	0.005	0.004	0.004
2.9	0.031	0.022	0.017	0.014	0.011	0.010	0.008	0.007	0.005	0.005	0.004	0.004	0.004	0.003	0.003
3.0	0.029	0.020	0.015	0.012	0.010	0.009	0.007	0.006	0.004	0.004	0.003	0.003	0.003	0.002	0.002
3.1	0.027	0.018	0.013	0.011	0.009	0.007	0.006	0.005	0.004	0.003	0.003	0.002	0.002	0.002	0.002
3.2	0.025	0.016	0.012	0.009	0.008	0.006	0.005	0.004	0.003	0.002	0.002	0.002	0.002	0.001	0.001
3.3	0.023	0.015	0.011	0.008	0.007	0.005	0.004	0.003	0.002	0.002	0.002	0.001	0.001	0.001	0.001
3.4	0.021	0.014	0.010	0.007	0.006	0.005	0.003	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001
3.5	0.020	0.012	0.009	0.006	0.005	0.004	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.001
3.6	0.018	0.011	0.008	0.006	0.004	0.004	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0+	0+
3.7	0.017	0.010	0.007	0.005	0.004	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0+	0+	0+
3.8	0.016	0.010	0.006	0.004	0.003	0.003	0.002	0.001	0.001	0.001	0.001	0+	0+	0+	0+
3.9	0.015	0.009	0.006	0.004	0.003	0.002	0.001	0.001	0.001	0.001	0+	0+	0+	0+	0+
4.0	0.014	0.008	0.005	0.004	0.003	0.002	0.001	0.001	0.001	0+	0+	0+	0+	0+	0+

For specific details about using this table to find *p*-values, see pages 484, 486.

P-VALUE FORMULAS IN Z-TEST

$$P=\begin{cases} P=2\times P([\Phi(|t_0|)]) & if \quad H_1:\mu\neq\mu_0 \\ P=1-\Phi(|t_0|)\ if\ t_0>0\ \&\ P=\Phi(|t_0|)\ if\ t_0<0 & if \quad H_1:\mu<\mu_0 \\ P=\Phi(|t_0|)\ if\ t_0>0\ \&\ P=1-\Phi(|t_0|)\ if\ t_0<0 & if \quad H_1:\mu>\mu_0 \end{cases}$$

EXAMPLES OF INFERENCE ABOUT MEAN WHEN SIGMA IS UNKNOWN

EXAMPLE-1 A medical investigation claims that the average number of infections per week at a hospital in southwestern Pennsylvania is 16.3. A random sample of 10 weeks had a mean number of 17.7 infections. The sample standard deviation is 1.8. Is there enough evidence to reject the investigator's claim at $\alpha = 0.05$?

SOLUTION *Here* $n = 10,\ \bar{x} = 17.7,\ s = 1.8,\ \alpha = 0.05$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu = 16.3$

$H_1 : \mu \neq 16.3$

Step -2: *Level of Significance*

$\alpha = 0.05$

Step -3: *Test Statistic*

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{17.7 - 16.3}{1.8/\sqrt{10}} = 2.46$$

Step -4: *Critical* Re *gion*

$$|t| \geq t_{\frac{\alpha}{2}(n-1)} \quad \Rightarrow |t| \geq t_{(0.025)(9)} \Rightarrow |t| \geq 2.262$$

Step -5: *Conclusion*

Since calculated value of z lies in CR so reject H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu = 16.3$

$H_1 : \mu \neq 16.3$

Step -2: *Level of Significance*

$\alpha = 0.05$

Step -3: *Test Statistic*

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{17.7 - 16.3}{1.8/\sqrt{10}} = 2.46$$

Step -4: $P-$*value*

$$p = 2[\Phi(|t_0|)] = 2[\Phi(|2.46|)] = 0.032 < P < 0.036$$

Step -5: *Conclusion*

Since $p-value < \alpha$ *so reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\bar{x} - t_{\frac{\alpha}{2}(n-1)} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(n-1)} \cdot \frac{s}{\sqrt{n}}$$

$$17.7 - (2.306)(1.8)/\sqrt{10} \leq \mu \leq 17.7 + (2.306)(1.8)/\sqrt{10}$$

$$16.39 \leq \mu \leq 19.01$$

Step -2: *Conclusion*

Since $\mu = 16.3$ *does not lie in CI so reject* H_0.

HAZBER SAMSON

EXAMPLE-2 An educator claims that the average salary of substitute teachers in school districts in New York, is less than \$60 per day. A random sample of eight school districts is selected, and the daily salaries (in dollars) are shown. Is there enough evidence to support the educator's claim at $\alpha = 0.10$?

60 56 60 55 70 55 60 55

SOLUTION *Here* $n = 8,\ \bar{x} = \sum x / n = 471/8 = 58.88,\ s = \sqrt{\dfrac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\dfrac{180.875}{7}} = 5.08,\ \alpha = 0.10$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu \geq \$60$

$H_1 : \mu < \$60$

Step -2: *Level of Significance*

$\alpha = 0.10$

Step -3: *Test Statistic*

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{58.88 - 60}{5.08/\sqrt{8}} = -0.624$$

Step -4: *Critical* Re *gion*

$t < -t_{\alpha(n-1)} \quad \Rightarrow t < -t_{(0.10)(7)} \Rightarrow t < -1.415$

Step -5: *Conclusion*

Since calculated value of t *does not lie in CR so do not reject* H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$H_0 : \mu \geq \$60$

$H_1 : \mu < \$60$

Step -2: *Level of Significance*

$\alpha = 0.10$

Step -3: *Test Statistic*

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{58.88 - 60}{5.08/\sqrt{8}} = -0.624$$

Step -4: $P - value$

$p = \Phi(|t_0|) = \Phi(0.624) = 0.284$

Step -5: *Conclusion*

Since $p - value > \alpha$ *so do not reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\mu \leq \bar{x} + t_{\alpha(n-1)} \cdot \frac{s}{\sqrt{n}}$$

$\mu \leq 58.88 + (1.415)(5.08)/\sqrt{8}$

$\mu \leq 61.42$

Step -2: *Conclusion*

Since $\mu = 60$ *lie in CI so do not reject* H_0.

HAZBER SAMSON

EXAMPLE-3 The Environmental Protection Agency (EPA) was suing the city of Rochester for noncompliance with carbon monoxide standards. EPA claimed that the mean level of carbon monoxide in downtown Rochester's air is dangerously high, higher than 4.9 parts per million. Does a random sample of 22 readings having sample mean 5.1 and sample standard deviation 1.17, present sufficient evidence to support the EPA's claim? Use $\alpha = 0.05$.

SOLUTION *Here* $n = 22,\ \bar{x} = 5.1,\ s = 1.17,\ \alpha = 0.05$

CRITICAL VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$$H_0 : \mu \le 4.9$$

$$H_1 : \mu > 4.9$$

Step -2: *Level of Significance*

$$\alpha = 0.05$$

Step -3: *Test Statistic*

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{5.1 - 4.9}{1.17/\sqrt{22}} = 0.80$$

Step -4: *Critical* Re *gion*

$$t > t_{\alpha(n-1)} \quad \Rightarrow t > t_{(0.05)(21)} \quad \Rightarrow t > 1.721$$

Step -5: *Conclusion*

Since calculated value of z does not lie in CR so do not reject H_0.

P-VALUE APPROACH

Step -1: *Formulation of Hypotheses*

$$H_0 : \mu \le 4.9$$

$$H_1 : \mu > 4.9$$

Step -2: *Level of Significance*

$$\alpha = 0.05$$

Step -3: *Test Statistic*

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{5.1 - 4.9}{1.17/\sqrt{22}} = 0.80$$

Step -4: $P - value$

$$p = \Phi(|t_0|) = \Phi(0.80) = 0.216$$

Step -5: *Conclusion*

Since $p - value > \alpha$ *so do not reject* H_0.

CONFIDENCE INTERVAL APPROACH

Step -1: *Formulation of Confidence Interval*

The $(1-\alpha)100\%$ *CI for* μ *is given by*

$$\mu \ge \bar{x} - t_{\alpha(n-1)} \cdot \frac{s}{\sqrt{n}}$$

$$\mu \ge 5.1 - (1.721)(1.17)/\sqrt{22}$$

$$\mu \ge 4.67$$

Step -2: *Conclusion*

Since $\mu = 4.9$ *lies in CI so do not reject* H_0.

Result Summary: At the 0.05 level of significance, the EPA does not have sufficient evidence to show that the mean carbon monoxide level is higher than 4.9.

HAZBER SAMSON